

Ayush Raj

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SKILLS

- **Languages:** C++, Python, C
- **Tools:** MS Office
- **Soft Skills:** Problem-Solving Skills, Team Player, Project Management, Adaptability

INTERNSHIP

Summer Intern

India Glycols Limited – Pharma City, Dehradun

July 2026

- Studied the extraction, biotransformation, purification, and finishing processes involved in the industrial production of thiocolchicoside from *Gloriosa superba* seeds.
- Gained practical exposure to upstream fermentation/bioconversion and downstream purification operations, along with the instruments and unit operations used in pharmaceutical manufacturing.
- Studied the upstream bioconversion process, including media preparation, sterilization, inoculum preparation, fermentation, and controlled conversion of thiocolchicine to thiocolchicoside.
- Learned about downstream processing involving column chromatography, solvent recovery, concentration using a wiped-film evaporator, crystallization, filtration, centrifugation, drying, and sieving.
- Observed the use of pH meters, TLC, UV-Vis spectrophotometry, centrifuges, reactors, filtration systems, and drying equipment for process monitoring and product recovery.
- Gained understanding of phytopharmaceutical extraction and purification, pharmaceutical-grade product processing, and plant-level quality requirements.

PROJECTS

Capstone Project | Study of Dengue in Female Anopheles Mosquitoes

December 2024

- Conducted a study on dengue and its interaction with female *Anopheles* mosquitoes, focusing on understanding why the mosquito is not adversely affected by dengue infection.
- Collected, organized, and analyzed relevant information to study mosquito–virus interactions and disease transmission mechanisms.
- Reviewed biological factors involved in dengue infection and mosquito–virus interactions, and interpreted research findings to understand mosquito vector biology and virus tolerance.

Arduino-Based Staircase Light System

October 2024

- Developed an Arduino-based automatic staircase lighting system using IR/motion sensors and LEDs for energy-efficient smart lighting automation.
- Integrated sensors and LEDs to enable automatic detection and illumination based on movement.
- Gained hands-on experience in embedded systems, sensor interfacing, Arduino programming, and basic circuit design.

Microbiology Game – Interactive HTML5 Game

September 2024

- Developed an interactive HTML5 Canvas-based educational game that combines microbiology concepts with engaging gameplay.
- Players control a friendly microbe, collect nutrients, avoid antibiotics, and progress through multiple evolutionary stages.
- The game incorporates timed microbiology quizzes every 20 seconds, rewarding correct responses with bonus points.
- Designed with responsive controls, accessibility features, scoring mechanics, and progressive difficulty to provide an engaging and educational user experience across desktop and mobile platforms.

CERTIFICATES

- Whole Genome Sequencing of Bacterial Genome – Coursera November 2025
- Introduction to Python – for Beginners – Udemey October 2024
- Hone Communication & Advance English Speaking Course – LPU September 2024

ACHIEVEMENTS

- Participated in an Idea pitching competition and got 2nd rank at Lovely Professional University. March 2026
- Successfully built and deployed a microbiology game and showcased on GitHub. September 2025
- github.com/ayush-raj05

EDUCATION

- **Lovely Professional University** Punjab, India
Bachelor of Technology – Biotechnology; CGPA: 6.5 August 2024 – July 2028
- **Progressive Central High School** Samastipur, Bihar
Intermediate; Percentage: 71% June 2020 – May 2022
- **Saraswati Shishu Mandir** Samastipur, Bihar
Matriculation; Percentage: 69.8% April 2019 – March 2020